



CSE 200 • B2 • Group 2

VARIATIONAL AUTOENCODER

Learning to generate, not just compress

Presented by **Member 1** • **Member 2** • **Member 3**

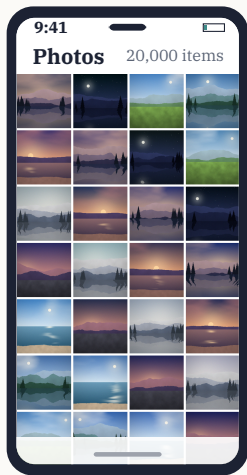
Department of CSE, BUET



latent space $z \sim \mathcal{N}(\mu, \sigma^2)$



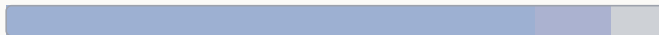
The Problem: Storage Almost Full



20,000 photos

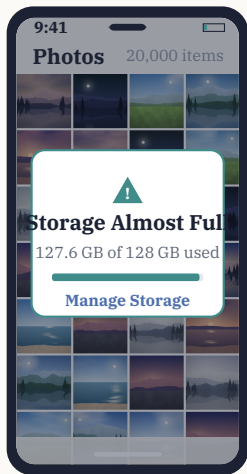
Imagine your phone has **20,000 photos**, but the storage is **almost full**.

Storage 127.6 GB of 128 GB used



■ Photos 112 GB ■ Apps 10 GB ■ System 5.6 GB

The Problem: Storage Almost Full



20,000 photos

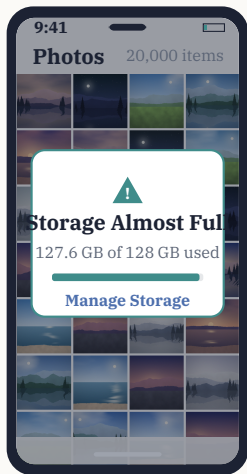
Imagine your phone has **20,000 photos**, but the storage is **almost full**.

Storage 127.6 GB used **only 0.4 GB free**



■ Photos 112 GB ■ Apps 10 GB ■ System 5.6 GB

The Problem: Storage Almost Full



20,000 photos

Imagine your phone has **20,000 photos**, but the storage is **almost full**.

Storage 127.6 GB used **only 0.4 GB free**



■ Photos 112 GB ■ Apps 10 GB ■ System 5.6 GB

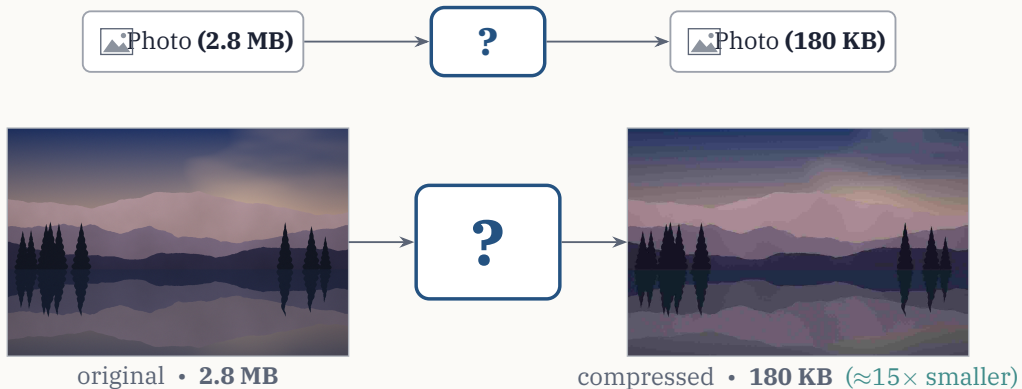
Delete memories?
✗ not ideal: once gone,
they're gone forever

A smarter solution
✓ keep every photo, but
use far less space

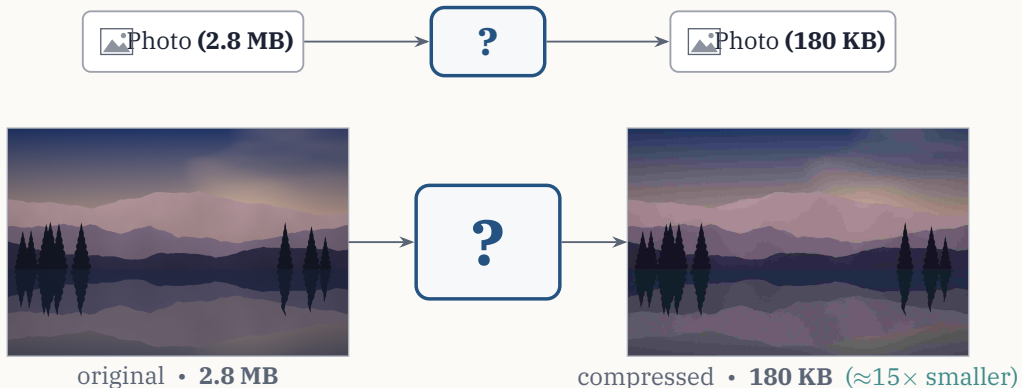
Compression Saves Storage



Compression Saves Storage



Compression Saves Storage

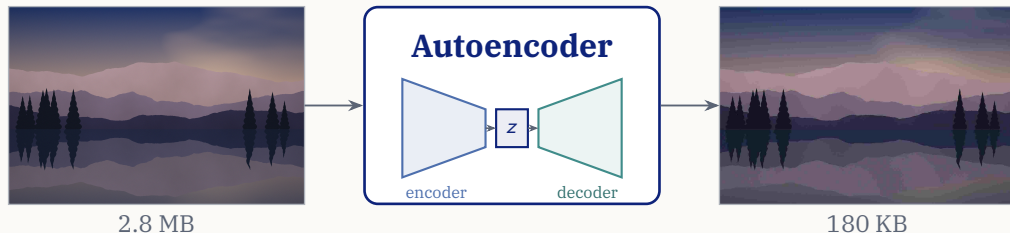


Same photo, much less storage. What goes inside the **mysterious middle box**?

Enter the Autoencoder

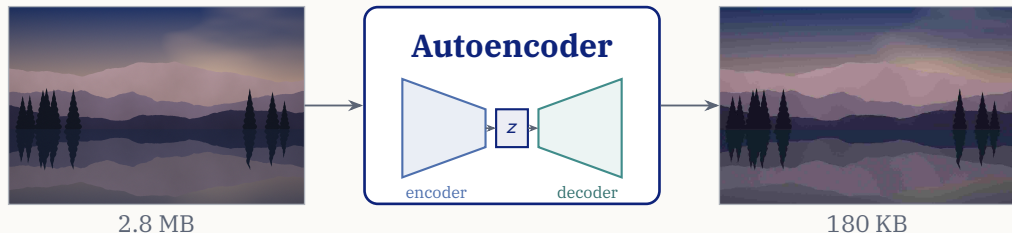


Enter the Autoencoder



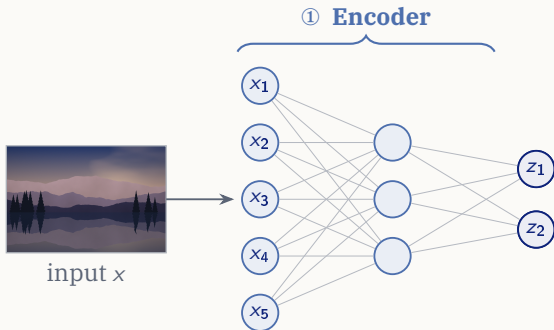
This box is exactly where **autoencoders** come in.

Enter the Autoencoder



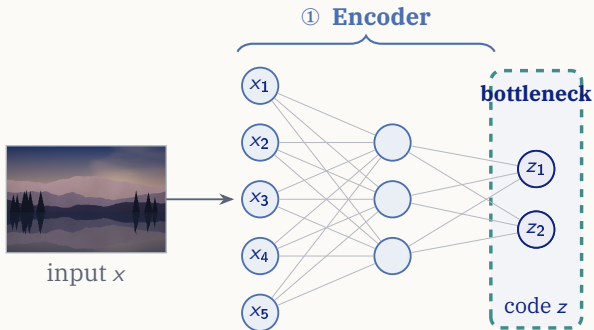
This box is exactly where **autoencoders** come in.
a neural network that learns to **compress** an image and **rebuild** it

Autoencoder Architecture



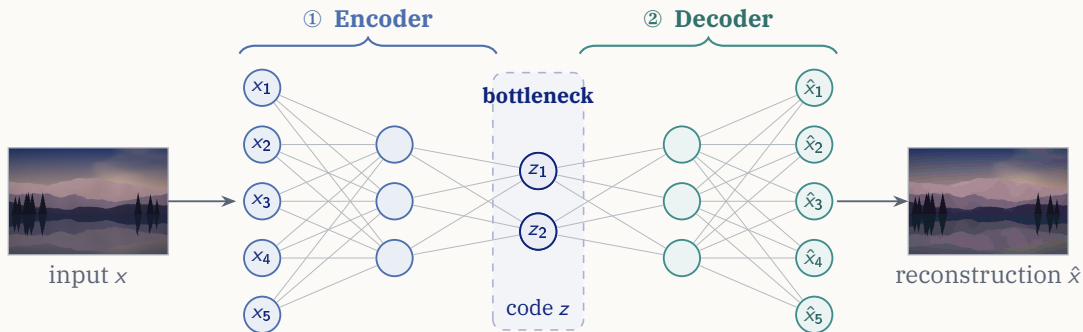
The **encoder** compresses the image, layer by layer, until it reaches the bottleneck.

Autoencoder Architecture



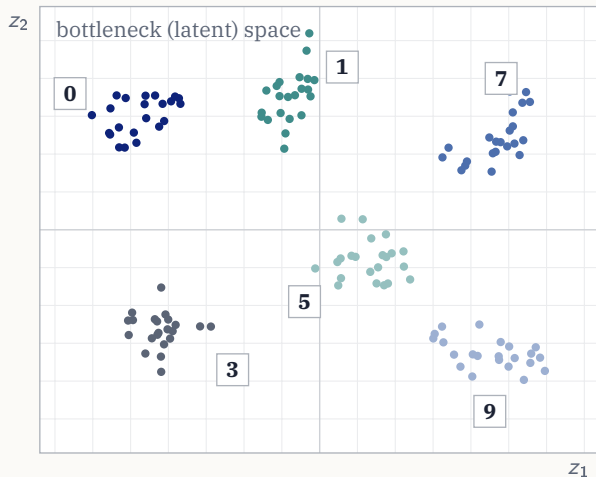
The **bottleneck** is a narrow doorway: only the *essential, non-redundant* information can pass.

Autoencoder Architecture



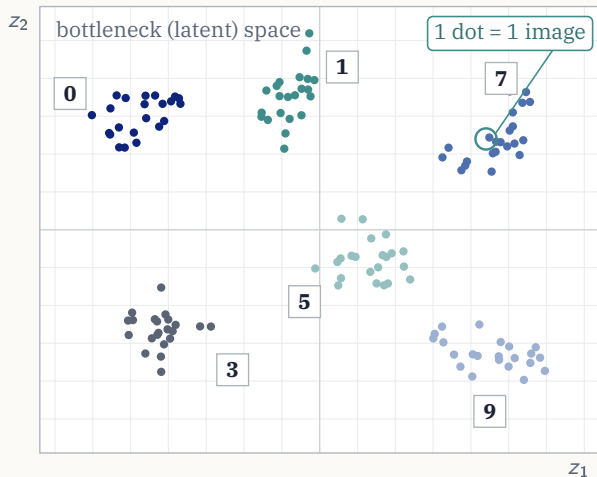
The **decoder** uses that compact code to rebuild the photo: $\hat{x} \approx x$.

The Problem with Autoencoders



Each point is **one image** after the autoencoder has compressed it to the bottleneck.

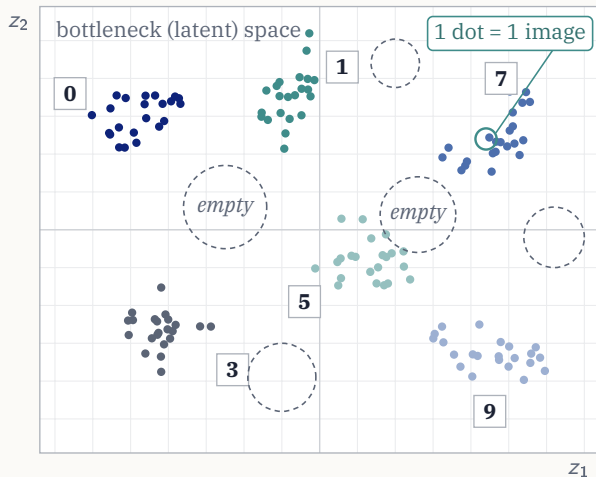
The Problem with Autoencoders



Each point is **one image** after the autoencoder has compressed it to the bottleneck.

Similar images land close together: the points form separate **islands**.

The Problem with Autoencoders

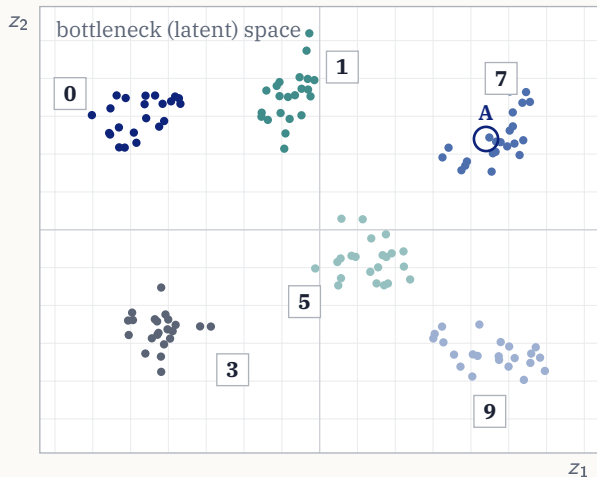


Each point is **one image** after the autoencoder has compressed it to the bottleneck.

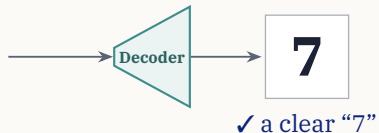
Similar images land close together: the points form separate **islands**.

Large **empty gaps** remain between the islands. The decoder has not seen training examples from these regions.

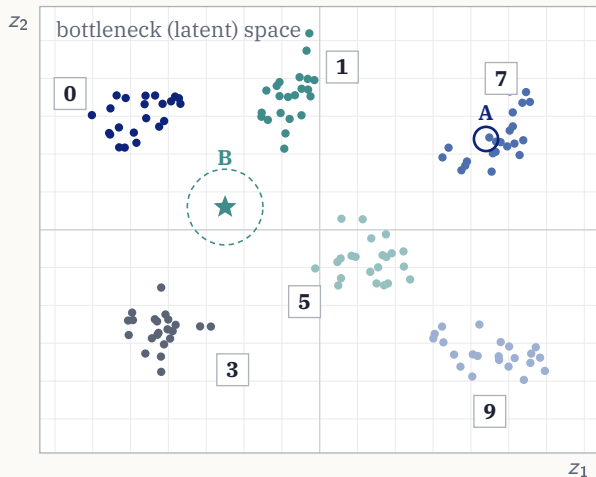
What Lives in the Gaps?



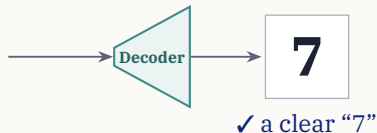
A point inside an island



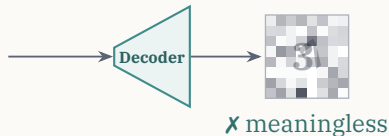
What Lives in the Gaps?



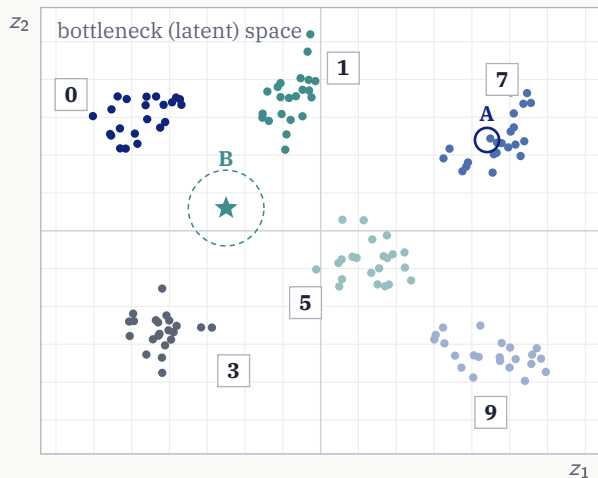
A point inside an island



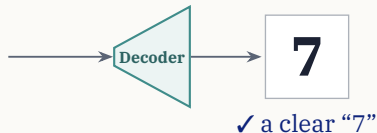
B random point from an empty gap



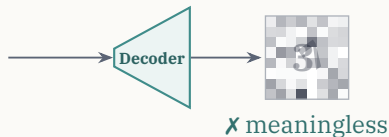
What Lives in the Gaps?



A point inside an island



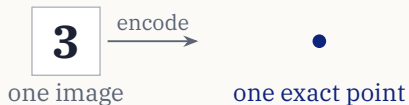
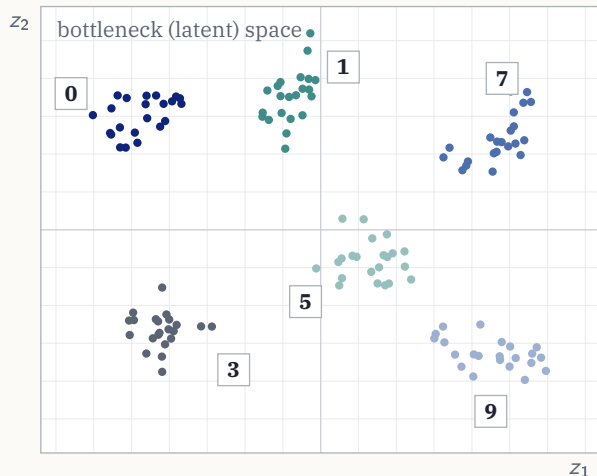
B random point from an empty gap



Suppose the encoder maps each image to a **distribution** instead of one point.

That is the central idea behind a VAE.

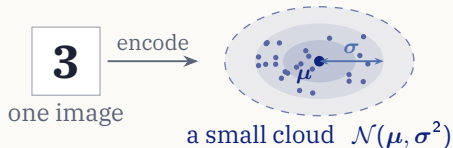
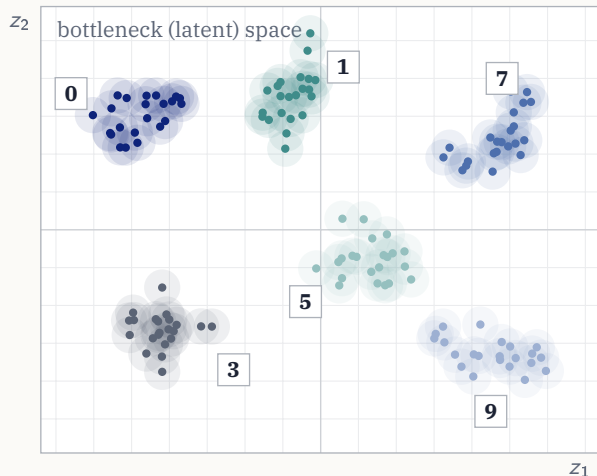
From Points to Distributions



Autoencoder: one image $\rightarrow$ one point

Right now every image is a single point, with empty gaps in between.

From Points to Distributions

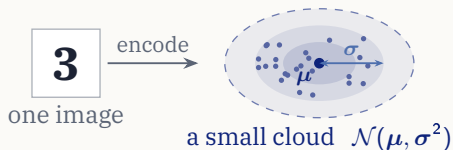
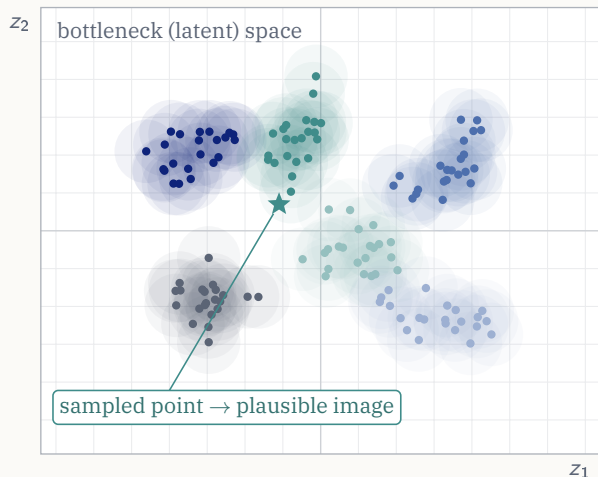


Autoencoder: one image $\rightarrow$ one point

VAE: one image $\rightarrow$ one **distribution**

Each image now occupies a **small area** instead of a single point.

From Points to Distributions

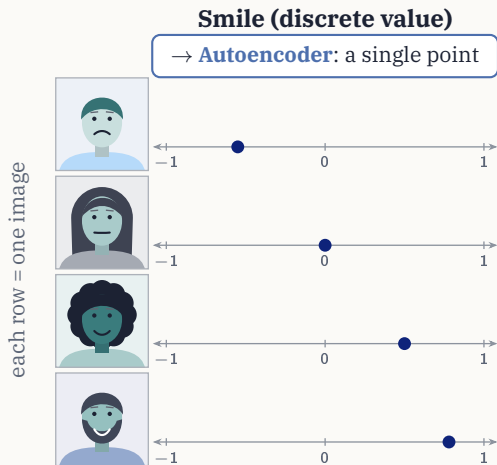


Autoencoder: one image $\rightarrow$ one point

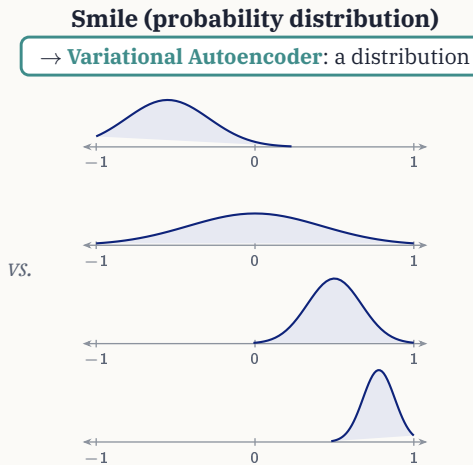
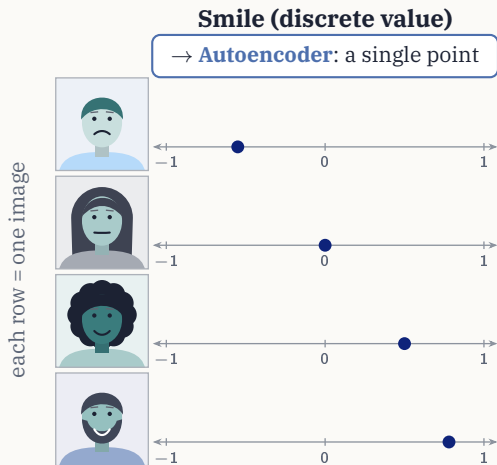
VAE: one image $\rightarrow$ one **distribution**

The distributions overlap, which **reduces empty gaps**. Nearby samples are more likely to decode into realistic images.

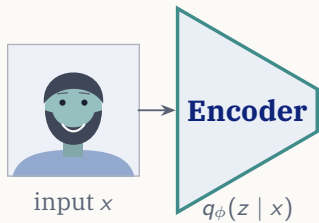
One Image, One Distribution



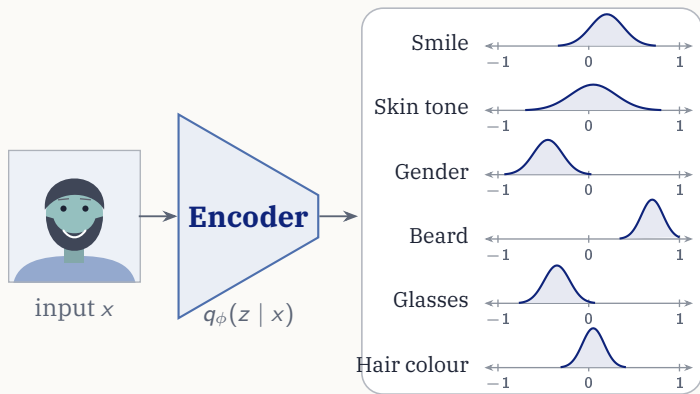
One Image, One Distribution



VAE Simplified: Latent Attributes

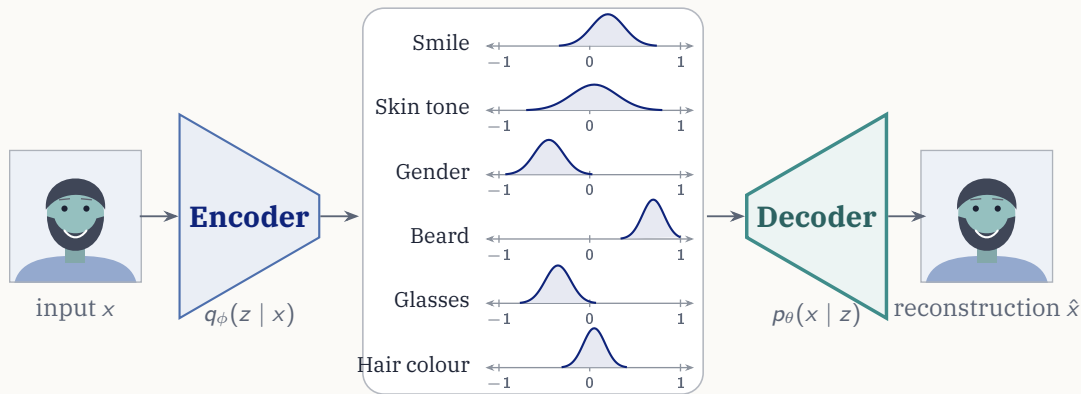


VAE Simplified: Latent Attributes



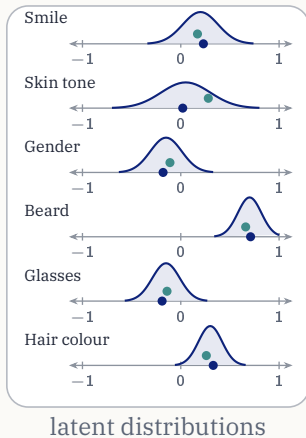
illustrative latent attributes: one distribution each

VAE Simplified: Latent Attributes

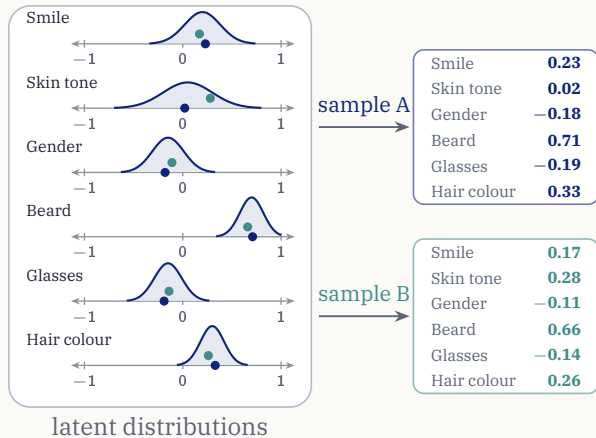


illustrative latent attributes: one distribution each

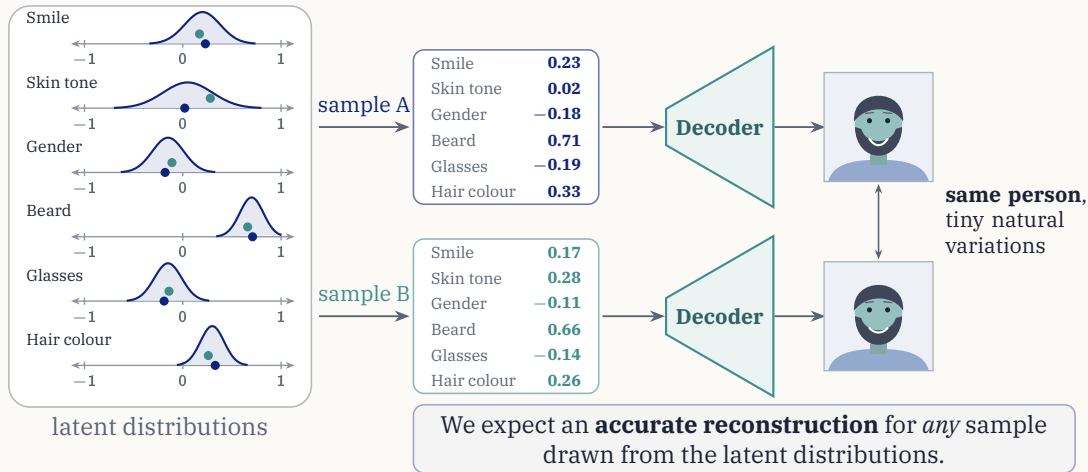
VAE Simplified: Sampling and Decoding



VAE Simplified: Sampling and Decoding

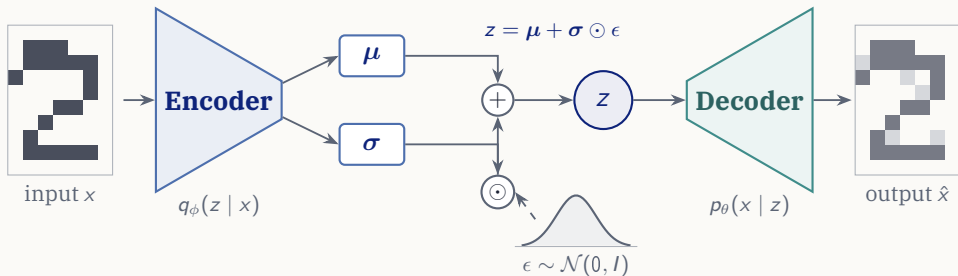


VAE Simplified: Sampling and Decoding



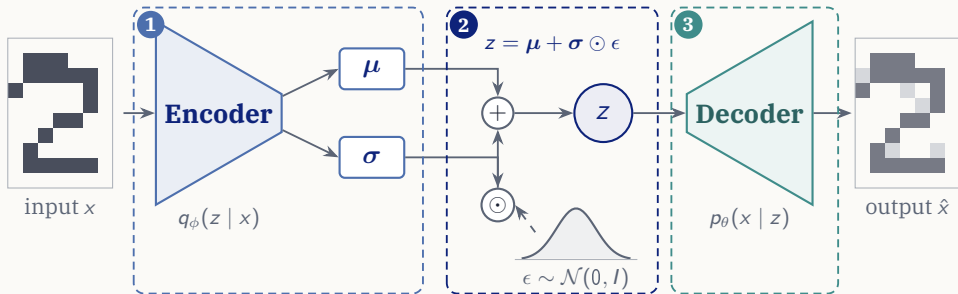
The VAE Workflow

So far: **Autoencoder** = one point per image • **VAE** = one distribution per image



The VAE Workflow

So far: **Autoencoder** = one point per image • **VAE** = one distribution per image



Handing over to **Member 2**: ① encoder $\rightarrow$ ② sampling $z \rightarrow$ ③ decoder